THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\tan 35^\circ \cdot \tan 55^\circ =$ 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 35

2. The value of $\sin 22^\circ / \cos 68^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 22

3. $(3\sin\theta)^2 + (3\cos\theta)^2 =$ 1

(a) 6 **(b)** 3 **(c)** 1 **(d)** 9

4. If $\sec\theta = \frac{13}{5}$ then $\cos\theta =$ 1

(a) $\frac{12}{13}$ **(b)** $\frac{5}{13}$ **(c)** $\frac{13}{5}$ **(d)** $\frac{12}{5}$

5. If $\tan A = \frac{15}{8}$, then $\sin A =$ 1

(a) $\frac{8}{17}$ **(b)** $\frac{15}{8}$ **(c)** $\frac{17}{15}$ **(d)** $\frac{15}{17}$

6. If $\cos A = \frac{7}{25}$, then $\tan A =$ 1

(a) $\frac{25}{7}$ **(b)** $\frac{24}{25}$ **(c)** $\frac{24}{7}$ **(d)** $\frac{7}{24}$

7. The value of $\sin^2 45^\circ + \cos^2 45^\circ$ is: 1

(a) 2 **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** $\frac{3}{2}$

8. $\tan 50^\circ \cdot \tan 40^\circ =$ 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 50

9. The value of $\sin 37^\circ / \cos 53^\circ$ is: 1

(a) 37 **(b)** 0 **(c)** 1 **(d)** 2

10. $(4\sin\theta)^2 + (4\cos\theta)^2 =$ 1

(a) 4 **(b)** 16 **(c)** 8 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Evaluate: $\dfrac{\sin 18^\circ}{\cos 72^\circ} + \dfrac{\tan 35^\circ}{\cot 55^\circ}$. 2

12. Evaluate: $\dfrac{\tan 33^\circ}{\cot 57^\circ}$. 2

13. If $\tan A = \frac{15}{8}$, find $\sec A$. 2

14. Evaluate: $\dfrac{\sin 32^\circ}{\cos 58^\circ} + \dfrac{\tan 50^\circ}{\cot 40^\circ}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $\tan\theta = \dfrac{24}{7}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

16. In $\triangle ABC$, right-angled at B, $AB = 9$ cm and $BC = 40$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

17. Prove that: $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. 3

OR

If $\tan\theta = \dfrac{21}{20}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

18. In $\triangle ABC$, right-angled at B, $AB = 6$ cm and $BC = 8$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

OR

Prove that: $\sec^2\theta + \csc^2\theta = \sec^2\theta\,\csc^2\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that: $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\csc\theta$. 5

OR

If $\tan A = \dfrac{15}{8}$, evaluate $\dfrac{2\sin A + 3\cos A}{5\sin A + 1\cos A}$. 5

20. Prove that: $(\sin\theta + \csc\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. 5

OR

If $\tan A = \dfrac{24}{7}$, evaluate $\dfrac{3\sin A + 1\cos A}{2\sin A + 1\cos A}$. 5

END OF QUESTION PAPER — Total: 40 Marks

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